

# Software Requirements Specification

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**Project Name:** Vendora

**Target Audience:** shop owners

**Version:** 1.0

**Date:** 10/7/2026

## 1. Introduction

### 1.1 Purpose:

The purpose of this document is to present a detailed description of the Vendora website, explaining its purpose and features.

This system is designed to enable shop owners to order company products electronically, replacing traditional method.

### 1.2 Problem Statement

Vendora distributes its products to retail shops through its sales representatives. However, the current manual management of this process has led to several fundamental issues.

The absence of a pre-ordering system leads to:

- **1. Wasted Time for Delivery Representatives:** Representatives must physically visit every shop just to ask them about the products they want to order from the company.
- **2. Product Availability Issues:** Representatives may run out of certain products during their distribution rounds.

### 1.3 Scope of Project

The project consists of a website that includes an administrative dashboard and an interface that enables shop owners to do the following:

- **A flexible product pre-ordering system.**
- **A flexible system for modifying/canceling orders.**
- **Seamless and flexible browsing of the company's products.**
- **An instant notification system for both shop owners and delivery representatives.**

## 2. Overall Description

## 2.1 User Classes:

- Shop owners
- Company Management

## 3. User Stories

**User story 1:** As a registered user (Shop Owner / Admin)

**I want to** log in to my account using my email and password,

**So that** I can securely access my dashboard and use the application features

### Acceptance Criteria:

1. **Success:** System redirects the user to their respective home dashboard upon entering valid credentials.
2. **Security:** Passwords must be masked during entry and encrypted during transit.
3. **Error Handling:** System displays a clear error message ("*Invalid email or password*") if wrong credentials are used.
4. **Password Recovery:** A "Forgot Password" link must be available to allow users to reset their credentials.

**User story 2:** As a shop owner,

**I want to** pre-order products through the website,

**So that** I can guarantee the availability and timely delivery of my requested items

### Acceptance Criteria:

- **Product Selection:** The shop owner can browse available products, specify quantities, and add them to a pre-order cart.
- **Order Summary:** Before final submission, the system must display an order summary including product names, quantities, unit prices, and the total estimated cost.
- **Confirmation:** Upon submitting the pre-order, the system generates a unique Order ID and changes the order status to "Pending/Received".
- **Notification:** The system sends an instant notification to both the shop owner (confirming receipt) and the delivery representative/management (notifying them of a new request)

### User story 3: As a shop owner,

**I want to** modify or cancel my placed orders through the website,

**So that** I have the flexibility to update my inventory needs based on sudden market or business changes

#### Acceptance Criteria:

- **Eligibility Window:** The shop owner can only modify or cancel an order while its status is "Pending" or "Processing" (before it is loaded for delivery).
- **Order Modification:** When modifying, the user can change product quantities or add/remove items. The system must recalculate the total cost instantly.
- **Order Cancellation:** When canceling, the system must ask for confirmation, change the status to "Canceled", and release any reserved stock back to the inventory.
- **Real-time Sync:** Any modifications or cancellations must immediately update on the Company Management dashboard and the Delivery Representative's view

### User story 4: As an admin,

**I want to** create, read, update, and delete products in the system,

**So that** I can seamlessly and flexibly manage the company's product catalog and inventory

#### Acceptance Criteria:

1. **Create (Add Product):** The admin can add a new product by entering its name, description, price, category, image, and available stock quantity.
2. **Read (View Products):** The admin can view a list of all products with a search and filter system (by category, price, or stock status).
3. **Update (Edit Product):** The admin can update any product details (e.g., changing prices or updating stock levels). Changes must reflect instantly for shop owners.
4. **Delete (Remove Product):** The admin can delete a product. If the product is linked to active or past orders, the system should perform a "Soft Delete" (archive/hide it) instead of a hard delete to maintain database integrity and order history.

### User story 5: As an admin,

**I want the system to** automatically route and assign shop orders to the designated delivery representative based on the shop's geographic region,

**So that** we can maximize operational efficiency and speed up the delivery workflow

### Acceptance Criteria:

1. **Region Mapping:** The admin can link each delivery representative to one or more specific geographic regions/zones in the system.
2. **Auto-Assignment:** When a shop owner submits a pre-order, the system checks the shop's registered region and automatically assigns the order to the representative responsible for that zone.
3. **Instant Notification:** Once assigned, the representative receives an immediate push notification/alert on their dashboard with the new order and shop details.
4. **Status Tracking:** The order status automatically updates to "Assigned to [Rep Name]" on the admin dashboard for tracking purposes

### Edge Cases:

- **Unassigned Region:** What happens if a shop in a new or unassigned region places an order, and no delivery representative is assigned to that zone yet?
  - *System Action:* The order must be routed as a "Pending Assignment" status, and an instant alert should be triggered on the Admin dashboard to manually assign a representative.
- **Representative on Leave / Offline:** If the representative responsible for a specific region is on leave or unavailable, how will the orders be routed?
  - *System Action:* The system must provide an option for the Admin to temporarily deactivate a representative's active status and set an alternative/backup representative to automatically receive that region's orders.

### User story 6: As a delivery representative,

**I want to** receive automated messages containing detailed shop orders **via Telegram,**

**So that** I can instantly view new tasks, save time, and verify product availability without constantly opening the main application

### Acceptance Criteria:

- **Account Linking (Telegram Integration):**

The system must provide an option in the delivery representative's profile to link their phone number or Telegram ID with the system's official **Telegram Bot** to enable message delivery.

- **Message Content:**

The automated Telegram message must clearly include the following details: Shop Name, a clickable link to the shop's geographic location (Google Maps), the itemized list of products and quantities, and the total invoice amount.

- **Instant Trigger:**

The Telegram notification must be triggered immediately upon the shop owner clicking "Confirm Order," with a delivery latency not exceeding 2 seconds

**User story 7:** As a delivery representative,

**I want to** receive an instant notification whenever a shop owner modifies their order,

**So that** I can verify the updated items before delivery and ensure the newly requested products are available in my stock.

**Acceptance Criteria:**

1. **Real-time Alert:** The system sends an immediate notification to the assigned representative the moment a shop owner saves any changes to a pending order.
2. **Change Highlights:** The notification view must clearly highlight what was changed (e.g., "*Product X increased from 2 to 5*" or "*Product Y removed*").
3. **Stock Validation:** The system checks if the new total quantity exceeds the representative's current van stock; if it does, a warning alert is displayed to both the representative and the shop owner

## **4.Functional Requirements:**

- **FR-1: User Authentication & Authorization**

The system must validate user credentials and provide secure, role-based login access for Shop Owners, Delivery Representatives, and Company Management.

- **FR-2: Pre-Order Management**

The system must allow registered shop owners to place product pre-orders, with the ability to modify or cancel their orders before shipping.

- **FR-3: Product Catalog Management (CRUD)**

The system must allow the Admin/Management to create, read, update, and delete products within the platform's catalog.

- **FR-4: Automated Order Routing via Telegram:** Upon pre-order submission, the system must automatically check the shop's geographic region, assign the order to the designated delivery representative, and send an **instant automated notification with full order details directly to the representative's Telegram account via Telegram API**

- **FR-5: Real-Time Notifications**

The system must send instant push notifications to the assigned delivery representative whenever a shop owner modifies or cancels their order.

- **FR-6: UI Responsiveness & Usability**

The system must provide flexible, user-friendly, and responsive interfaces tailored to the operational needs of Shop Owners, Delivery Reps, and Admins

## **5.Non Functional Requirements:**

- **NFR-1: Security**

All user passwords and sensitive financial data must be securely encrypted using robust hashing algorithms. The system must be fully protected against common security vulnerabilities and attacks.

- **NFR-2: Performance & Speed**

Core operations, including product catalog loading, pre-order submissions, and automated Telegram message routing, must execute within a maximum response time of 2 to 3 seconds under normal network conditions.

- **NFR-3: Availability & Reliability**

The system must guarantee an uptime stability of at least 99% daily, specifically from 8:00 AM to 1:00 PM, ensuring uninterrupted service during peak delivery hours and operation shifts.

- **NFR-4: Usability**

The delivery representative's mobile interface must be highly intuitive, minimal, and optimized for fast, safe usage during transit. The UI must be fully responsive across various device screens using a flexible utility-first framework like Tailwind CSS.

- **NFR-5: Scalability**

The system architecture and the relational database management system (such as MySQL) must handle horizontal growth smoothly, sustaining an increasing volume of registered shops, daily pre-orders, and delivery reps without performance degradation.

- **NFR-6: Compatibility**

The web application must be fully compatible and error-free across all major modern web browsers, including Google Chrome, Microsoft Edge, Safari, and Mozilla Firefox

## 6. Tech Space

### 6.1. Database Design

*This defines the core tables, their attributes, and relationships.*

#### users (System Accounts)

- **id**: Primary Key
- **name**: String
- **email**: String (Unique)
- **password**: String (Hashed)
- **role**: Enum (*admin, shop\_owner, delivery\_rep*)
- **telegram\_id**: String (Nullable)
- **created\_at**: Timestamp

#### regions (Geographic Distribution)

- **id**: Primary Key
- **name**: String
- **delivery\_rep\_id**: Foreign Key (references users.id)

#### shop\_details (Retailer Logistical Profiles)

- **id**: Primary Key
- **user\_id**: Foreign Key (references users.id, Unique)
- **shop\_name**: String
- **region\_id**: Foreign Key (references regions.id)
- **location\_gps**: Text / String (Google Maps Link)
- **created\_at**: Timestamp

## products (Company Inventory Items)

- **id**: Primary Key
- **name**: String
- **description**: Text (Nullable)
- **number\_of\_pieces**: Integer
- **size\_or\_weight**: String (Nullable)
- **price**: Decimal
- **stock\_quantity**: Integer (Default: 0)
- **image\_path**: String (Nullable)
- **created\_at**: Timestamp

## orders (Pre-Order Submissions)

- **id**: Primary Key
- **shop\_owner\_id**: Foreign Key (references users.id)
- **delivery\_rep\_id**: Foreign Key (references users.id, Nullable)
- **total\_price**: Decimal
- **status**: Enum (*pending*, *processing*, *assigned*, *canceled*)
- **created\_at**: Timestamp

## order\_items (Order Line Details / Pivot)

- **id**: Primary Key
- **order\_id**: Foreign Key (references orders.id)
- **product\_id**: Foreign Key (references products.id)
- **quantity**: Integer
- **price**: Decimal

